

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	N. Poanifa	Reg. No.:	192525322
Section / Batch:	2025 - 2029	Date:	20.7.26

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions	Marks
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:			100 Marks

SECTION A – CONCEPT MAPPING

- 1) Selection and Bubble Sort
- 2) Master Theorem

Code of Conduct

I N. Parvita (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: N. Parvita

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks (Each – 50)
Q1						
Q2						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

SELECTION SORT AND BUBBLE SORT

1. What are they?

- * Simple comparison-based sorting algorithms
- * Used to arrange elements in ascending order

4. Time complexity

- * Bubble sort is adaptive (best case $O(n)$ when already sorted)

5. Space complexity

- * Both use $O(1)$ extra space
- * They are in-place sorting algorithms

2. Good

- * Sort an array of n elements in increasing order.

6. Comparison

Selection sort	Bubble sort
* Fewer swaps (at most $n-1$)	* More swaps (in average)
* Writes are less	* Easy to implement

3. Key Idea

- * Selection sort: Repeatedly select the minimum element and place it in the correct position.
- * Bubble Sort: Repeatedly compare adjacent elements and swap if they are in wrong order.

7. Applications

- * Useful for small datasets.
- * Easy to understand and implement.
- * Rarely used for large datasets (higher time).

MASTER THEOREM

1. What is it

Master Theorem provides asymptotic bounds for divide-and-conquer recurrences of the form

$$T(n) = aT(n/b) + f(n)$$

where $a \geq 1, b > 1, f(n) > 0$

2. Purpose

Gives a quick way to find the asymptotic time complexity of divide-and-conquer algorithms

3. Parameters

a : number of subproblems

b : factor by which size is reduced

$f(n)$: work done outside recursion

$$T(1) = \Theta(1)$$

4. The Three Cases

Case 1

If $f(n) = O(n^{\log_b a - \epsilon})$
for some $\epsilon > 0$
 $T(n) = \Theta(n^{\log_b a})$
(subproblem cost dominates)

Case 2

If $f(n) = \Theta(n^{\log_b a} \log^k n)$
for some $k \geq 0$
 $T(n) = \Theta(n^{\log_b a} \log^{k+1} n)$
(Balanced case)

Case 3

If $f(n) = \Omega(n^{\log_b a + \epsilon})$
for some $\epsilon > 0$
 $T(n) = \Theta(f(n))$
(work outside dominates)
+ Regularity condition must hold